

Noncommutative Comparative Statics

A defect calculus for order-sensitive adjustment protocols

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Abstract

Many constrained systems do not simply jump to a uniquely selected optimum when their external conditions change. They carry their current state forward, make a locally economical repair, and retain the residue of earlier repairs. Two externally compatible interventions can therefore produce different outcomes, or different feasibility, when applied in different orders. We propose *noncommutative comparative statics* (NCS) as a mathematical research program centered on this phenomenon.

The basic model is a presented category of external interventions together with a response functor from its free path category to metric spaces and partial Lipschitz maps. A relation cell declares two intervention paths externally equivalent; the response need not satisfy that relation. We separate directional route-asymmetric failure, common failure, and finite value discrepancy. Signed-distance predicate robustness supplies a precise baseline: a finite route difference can become one-sided infeasibility under a later partial response only near the continuation domain’s feasibility boundary. We give its one-step two-sided boundary-exposure form and use it in a three-intervention example. We also define an amplitude-calibrated reporting order, which distinguishes the quadratic small cells of smooth connection curvature from linear active-set effects and order-zero jumps.

The paper imports, rather than relabels as new, quantitative rewriting, Hyers–Ulam rectification, predicate robustness, discrete connections, ordered projection, hybrid-system, and comparative-statics results. An affine-contractive relation operator gives a computable baseline rectification problem, including noninvertible maps. Exact configuration-repair and online-allocation examples show how the same definitions report finite order debt and hard feasibility asymmetry. The proposed contribution is an explicitly delimited shared modeling framework, falsification protocol, and research agenda—not a novelty claim for noncommutativity, curvature, margins, confluence, or sequential recourse.

Keywords. comparative statics; partial maps; adjustment cost; path dependence; quantitative rewriting; active sets; algorithmic recourse; Ulam stability.

Status and priority boundary. This is a proposal for a field and a first foundations paper, not a claim that the field is already established. A mapped literature search found no source packaging the full response/failure/scaling framework below, but every major mathematical ingredient has close antecedents. No theorem in this paper by itself establishes a separate field. The proposal is limited to the combined intervention/response schema, directional reporting convention, cross-sector falsification protocol, and research agenda.

1 The missing question

Classical comparative statics asks how an equilibrium or optimizer changes with an external parameter. This is exactly the right question when the model specifies a unique state $x^*(b)$ for every parameter b . It is not the right question for a stateful repair protocol.

Table 1: Classical and noncommutative comparative statics ask different questions.

	Classical comparative statics	NCS
State semantics	selected equilibrium $x^*(b)$	current state plus an adjustment protocol
Primary comparison	two external endpoints	two externally equivalent paths
Failure	often excluded by regularity	an observable distinct from value error
Path independence	built into a state function	a property to test or rectify
Local diagnostic	derivative of x^*	relation-cell response signature

Consider a system with a current feasible state. A requirement changes. The system minimizes disruption from *where it currently is*, rather than re-solving from a timeless reference. A second requirement then arrives and the system repairs again. Even if the final set of requirements is independent of their order, the accumulated state need not be.

This motif recurs in technical problem solving:

- a configuration manager preserves earlier edits while satisfying a new constraint;
- an allocation system admits arrivals without moving incumbents;
- a biological or engineered redundant system redistributes internal load as its environment changes;
- an active-set optimizer crosses different constraint faces in different orders;
- a person follows sequential algorithmic-recourse recommendations.

The pattern is especially salient in requests of the form “change this one thing while keeping everything else consistent.” It appears across learned problem-solving corpora and across fields, but each field gives it a different mathematical home. The proposal here is to make the *composition of response protocols* the primary object.¹

The under-formalized claim is therefore organizational and empirical, not that partial maps or noncommuting operators were unknown. Models commonly specify either the external endpoint, the internal dynamics, or the failure rule. NCS asks them to specify together (i) which intervention paths are externally equivalent, (ii) how current state is carried along each path, and (iii) which directional failure and value discrepancies are measured when the response does not respect that equivalence.

1.1 Reset and carry

The first conceptual distinction is simple and non-negotiable.

Reset response. Recompute a distinguished state from a fixed reference for every new external condition. A unique reset state is a section and is path-independent by construction.

Carry response. Repair from the current state. Earlier accommodations remain unless a later requirement forces them to change.

¹This is a hypothesis drawn from recurring problem structures, not an inspection of individual neural network weights.

Confusing these semantics creates a false theorem: a unique static argmin cannot acquire holonomy merely because its derivative is written geometrically. NCS treats reset and carry as different response functors on the same feasible fibers.

2 Position among established fields

The proposed field sits at an interface. This section records the collisions that determine what may and may not be claimed.

Comparative statics and recourse. Monotone comparative statics is a mature theory [Milgrom and Shannon, 1994]; adjustment costs have been incorporated explicitly [Dekel et al., 2025]; and “recourse” has long-standing meanings in stochastic programming [Dantzig, 1955]. Sequential and ordered algorithmic recourse is also established [Rasouli and Yu, 2024, Verma et al., 2022]. NCS does not claim that order-sensitive adjustment is undiscovered. It seeks a common defect language for the response maps those models generate.

Rewriting and concurrency. A pair of competing response paths is a quantitative diamond. Adjacent swaps of independent actions are the basis of trace theory [Mazurkiewicz, 1988], directed concurrency [Kahl, 2016, Earnshaw et al., 2026], and quantitative rewriting [Gavazzo and Di Florio, 2023]. Metric-enriched categories already express approximately commuting diagrams [Kubiś, 2013]. Accordingly, the triangle-inequality accumulation bound in section 5 is a baseline, not a headline result.

Connections and holonomy. Horizontal lift, curvature, and holonomy are classical [Ehresmann, 1952, Ambrose and Singer, 1953, O’Neill, 1966]. In redundant robotics, minimum-norm feasible velocity is explicitly the pseudoinverse horizontal lift [Altafini, 2004, Shamir and Yomdin, 1988]. Discrete edge transport and plaquette curvature are likewise established [Leok et al., 2005, Fernández et al., 2022, Wilson, 1974]. The smooth sector of NCS is a reduction to this theory.

Moving constraints and hybrid seams. Sweeping processes model motion constrained by a moving set [Moreau, 1977]; rate-independent systems model incremental dissipation and hysteresis [Mielke and Theil, 2004]; and projection algorithms already study ordered noninvertible operators [Bauschke and Borwein, 1996]. Exit-path categories and hybrid systems model noninvertible cross-stratum transport and saltation products [Treumann, 2009, Woolf, 2009, Burden et al., 2016, Kong et al., 2024]. Scattering diagrams organize path-ordered wall products and joint consistency [Gross et al., 2018]. NCS adds no canonical seam map to a stratification; the response protocol must supply it.

Predicate robustness and partial maps. Distance to a predicate boundary as a truth-preserving robustness radius is established in metric temporal logic and hybrid systems [Fainekos and Pappas, 2009]. Boundary-band risk is also standard in robust classification [Zhang et al., 2019], while convergence theories compare partial maps through both their domains and values [Beer et al., 2014]. Accordingly, the guard lemma in section 6 and its one-step boundary-exposure estimate are imported baselines. Their role is to connect those quantities to competing adjustment paths, not to rename them as new mathematics.

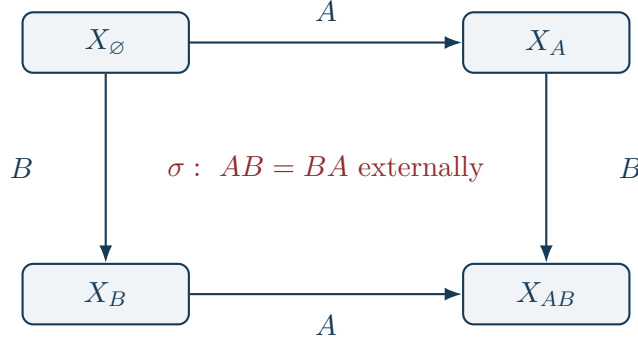


Figure 1: A relation cell equates two external intervention orders. The response maps on the four edges need not make the state square commute.

Rectification and Ulam stability. The question “does almost satisfying the relations imply proximity to an exact representation?” is Ulam stability [Burger et al., 2010, Gowers and Hatami, 2015, Voiculescu, 1983, Becker and Chapman, 2023, Arzhantseva and Păunescu, 2015]. For a Hilbert operator, the optimal residual-to-kernel error constant is the norm of its Moore–Penrose inverse [Huang and Moslehian, 2012]. We use this theorem as a computational baseline for an affine response sector; we do not present it as a new rectification theorem.

3 Presented interventions and response functors

3.1 External intervention syntax

Definition 3.1 (weighted intervention presentation). An intervention presentation

$$K = (K_0, K_1, K_2, \{p_\sigma \Rightarrow q_\sigma\}, w_1, w_2)$$

consists of a finite directed graph, a finite family of pairs of nonempty parallel paths, and positive edge and cell weights. The cell $p_\sigma \Rightarrow q_\sigma$ declares that the two paths have the same net external meaning.

Let $\text{Path}(K)$ be the free category on the graph and set

$$\mathcal{C}_K = \text{Path}(K) / \langle p_\sigma = q_\sigma : \sigma \in K_2 \rangle.$$

The quotient is the external intervention category. The presentation matters: it records which changes are elementary, which exchanges are operationally meaningful, and how evidence is weighted. It is not merely a redundant description of $\mathcal{C}_K$.

3.2 Partial Lipschitz response

Definition 3.2 (partial Lipschitz map). A partial Lipschitz map $T : X \rightarrow Y$ is a subset $\text{Dom } T \subseteq X$ and a map $T : \text{Dom } T \rightarrow Y$ for which some finite L satisfies

$$d_Y(Tx, Tx') \leq Ld_X(x, x')$$

whenever $d_X(x, x') < +\infty$. Its modulus $\text{Lip}(T)$ is the infimum of such L . Composition has domain

$$\text{Dom}(S \circ T) = \{x \in \text{Dom } T : T(x) \in \text{Dom } S\}.$$

These arrows form the category $\mathbf{pMetPar}_{\text{Lip}}$.

Definition 3.3 (response assignment). A deterministic response assignment is a functor

$$R : \text{Path}(K) \longrightarrow \mathbf{pMetPar}_{\text{Lip}}.$$

It is *exact* relative to K if it factors through $\mathcal{C}_K$.

The pair (K, R) is the basic *NCS model*. A statistical study also declares input measures and value scales; a scaling study additionally declares physical intervention amplitudes. These choices are part of the model, not canonical decorations.

The Lipschitz target is intentionally wider than nonexpansive response. A unique optimizer need not depend nonexpansively on its reference state. Smooth horizontal transports between fibers with changing metrics need not be nonexpansive either. The model must prove or record the relevant local Lipschitz modulus.

3.3 Morphisms and gauge

An *unweighted semantic morphism* $(F, \eta) : (K, R) \rightarrow (K', R')$ contains a functor $F : \text{Path}(K) \rightarrow \text{Path}(K')$ satisfying

$$F(p_\sigma) \sim_{K'} F(q_\sigma)$$

for every relation and a natural transformation $\eta : R \Rightarrow R'F$ with total finite-Lipschitz components in the partial-map category. Mapping an edge to a path permits operational refinement. This notion forgets w_1, w_2 . A quantitative morphism would also need chosen target derivations and bounds on edge-path and cell-weight distortion; we leave its canonical formulation open.

A gauge equivalence fixes the intervention presentation and uses fiberwise bijective isometries. It preserves route domains and value distances. Uniformly bi-Lipschitz fiber changes give two-sided distortion bounds. Presentation equivalence, quotient-category equivalence, and response gauge equivalence are different notions.

4 A failure-aware order signature

Let $p, q : a \rightarrow b$ be parallel intervention paths and let $D_p = \text{Dom } R_p$, $D_q = \text{Dom } R_q$.

Definition 4.1 (domain/value signature). The deterministic order signature is

$$\begin{aligned} A_{p,q}^+ &= D_p \setminus D_q, & A_{p,q}^- &= D_q \setminus D_p, \\ C_{p,q} &= X_a \setminus (D_p \cup D_q), & V_{p,q}(x) &= d_b(R_p x, R_q x), \quad x \in D_p \cap D_q. \end{aligned}$$

The four entries distinguish:

1. only the first order succeeds;
2. only the second order succeeds;

3. both orders fail;
4. both succeed but disagree in value.

A single extended distance that assigns $+\infty$ to one-sided failure and zero to common failure is convenient algebraically but scientifically incomplete. In particular, universal failure is path-independent and useless.

For a probability measure μ on X_a and a truncation scale $M > 0$, define the directional empirical signature

$$A_\mu^+(p, q) = \mu(D_p \setminus D_q), \quad (1)$$

$$A_\mu^-(p, q) = \mu(D_q \setminus D_p), \quad (2)$$

$$C_\mu(p, q) = \mu(\mathbb{C}_{p,q}), \quad (3)$$

$$V_{\mu,M}(p, q) = \int_{D_p \cap D_q} \min \left\{ 1, \frac{d_b(R_p x, R_q x)}{M} \right\} d\mu(x). \quad (4)$$

This produces the directly estimable order signature $(A_\mu^+, A_\mu^-, C_\mu, V_{\mu,M})$. Write

$$A_\mu^{\leftrightarrow} = A_\mu^+ + A_\mu^- = \mu(D_p \triangle D_q)$$

when only total asymmetry is needed. Retaining the two directions is essential: their sum cannot say which order fails.

This convention is useful rather than mathematically exotic. Totalize a partial map by an undefined state $\perp$, put $d_M^\perp(y, y') = \min\{1, d(y, y')/M\}$, and set $d_M^\perp(y, \perp) = 1$. Then

$$\int d_M^\perp(\hat{p}, \hat{q}) d\mu = A_\mu^{\leftrightarrow} + V_{\mu,M},$$

while C_μ is common- $\perp$ mass. Thus the signature is a scientifically interpretable decomposition of standard objects, not a claim of a new metric on partial maps.

Proposition 4.2 (exact descent). *The response R factors through $\mathcal{C}_K$ if and only if every relation cell has equal route domains and zero value defect on that domain.*

Proof. The condition is equality of the corresponding partial maps. The universal property of the presented category gives the factorization. $\square$

Common failure is compatible with exact descent, which is why it must be retained as a separate performance observable.

5 Baseline filling calculus

Suppose $P_0 = p, \dots, P_N = q$ is a relation-cell swap derivation at x and every intermediate route is defined at x . At step i , write the two routes as a common prefix a_i , alternative cell sides u_i, v_i , and a common suffix b_i . Choose a positive finite Lipschitz bound $L_{b_i} > 0$ for each suffix response. Using chosen positive bounds avoids the indeterminate product $0 \cdot +\infty$ allowed by extended metrics.

Proposition 5.1 (admissible derivation bound).

$$d(R_p x, R_q x) \leq \sum_{i=0}^{N-1} L_{b_i} d(R_{u_i} R_{a_i} x, R_{v_i} R_{a_i} x).$$

Proof. Use the triangle inequality across successive route outcomes and propagate each local comparison through its common suffix. $\square$

The response filling cost is the infimum of the right-hand side over *admissible* derivations and valid positive suffix bounds. It is $+\infty$ if none exists. Ordinary combinatorial filling area is safe only for total responses or after a domain-robustness theorem. This distinction is essential: a small local value difference can straddle the domain of a later partial map and become one-sided failure.

6 Imported guard robustness and boundary exposure

For a subset $D \subseteq Y$, define its membership margin at y by

$$m_D(y) = \begin{cases} \text{dist}(y, Y \setminus D), & y \in D, \\ \text{dist}(y, D), & y \notin D. \end{cases}$$

Distances to the empty set are $+\infty$.

Lemma 6.1 (signed-distance guard robustness). *Let $p, q : X \rightarrow Y$ be route responses and let $r : Y \rightarrow Z$ be a continuation with domain D and Lipschitz constant L . Fix $x \in \text{Dom } p \cap \text{Dom } q$, write*

$$y = p(x), \quad y' = q(x), \quad \delta = d_Y(y, y').$$

If $\delta < m_D(y)$, then $y \in D$ if and only if $y' \in D$. If both lie in D , then

$$d_Z(r(y), r(y')) \leq L\delta.$$

The same statement holds with $m_D(y')$ in place of $m_D(y)$.

Proof. If $y \in D$ and $y' \notin D$, then $\delta \geq \text{dist}(y, Y \setminus D) = m_D(y)$, a contradiction. If $y \notin D$ and $y' \in D$, then $\delta \geq \text{dist}(y, D) = m_D(y)$. Once both points lie in D , apply the Lipschitz inequality. $\square$

This is the absolute signed-distance robustness certificate for the predicate 1_D , already established in substantially richer temporal and hybrid settings [Fainekos and Pappas, 2009]. Its role here is structural: after substituting two route outcomes for y, y' , it identifies the quantity connecting finite path disagreement to hard feasibility disagreement.

Corollary 6.2 (one-step boundary exposure). *Let $E = \text{Dom } p \cap \text{Dom } q$ and $\delta(x) = d_Y(p(x), q(x))$. Assume the domains, maps, and margins below are measurable. Define*

$$S_\mu^\cap(p, q; r) = \mu \{x \in E : \max(m_D(p(x)), m_D(q(x))) \leq \delta(x)\}.$$

Then

$$A_\mu^{\leftrightarrow}(rp, rq) \leq A_\mu^{\leftrightarrow}(p, q) + S_\mu^\cap(p, q; r).$$

Proof. Partition inputs into original one-sided prefix failure, common prefix failure, and common prefix success. The first class is charged to $A_\mu^{\leftrightarrow}(p, q)$, while both composites fail in the second. In the third, continuation membership can differ only if each endpoint lies within their mutual distance of the opposite-membership set, so both margins are at most $\delta(x)$. $\square$

If $\delta(x) \leq \varepsilon$ on E , one obtains a looser uniform bound by replacing $\delta(x)$ with ε . The maximum is important: membership disagreement forces both route outcomes into their respective seam bands, not merely one. Distributionally, this is an indicator union bound related to boundary risk [Zhang et al., 2019]; it is a baseline, not a novelty theorem.

Example 6.3 (necessity of seam exposure). Let $p(*) = 0$, $q(*) = \varepsilon$, and let $\text{Dom } r = (-\infty, \varepsilon/2)$. The prefix value defect is only ε , yet exactly one continuation succeeds. Both prefix outcomes are within ε of the continuation seam.

The empirical implication is useful. Estimate local two-route disagreement, estimate how much response mass lies near a future feasibility boundary, and predict an upper bound on order-specific failure without fitting the longer paths themselves.

7 Affine-contractive rectification as a baseline

Let each vertex b carry a finite-dimensional Hilbert space H_b . Give each edge a fixed linear contraction

$$A_e : H_{s(e)} \rightarrow H_{t(e)}$$

and an adjustable bias $a_e \in H_{t(e)}$:

$$R_e^a(x) = A_e x + a_e.$$

The maps are total and nonexpansive, and A_e may be noninvertible.

Assume the linear background satisfies every relation:

$$A_{p\sigma} = A_{q\sigma}.$$

For $p = e_n \cdots e_1$, define

$$B_p(a) = a_{e_n} + A_{e_n} a_{e_{n-1}} + \cdots + A_{e_n} \cdots A_{e_2} a_{e_1}.$$

The affine relation operator

$$D_A : \bigoplus_{e \in K_1} H_{t(e)} \longrightarrow \bigoplus_{\sigma \in K_2} H_{t(p_\sigma)}$$

is

$$(D_A a)_\sigma = B_{p_\sigma}(a) - B_{q_\sigma}(a).$$

Exact affine response is equivalent to $D_A a = 0$.

Proposition 7.1 (affine Hyers–Ulam rectification). *The nearest exact affine bias in the declared Hilbert direct-sum norm is*

$$\bar{a} = (I - D_A^\dagger D_A) a,$$

and

$$\text{dist}(a, \ker D_A) \leq \|D_A^\dagger\| \|D_A a\|.$$

If $D_A \neq 0$, the optimal constant is $1/\sigma_{\min}^+(D_A)$.

Proof. This is orthogonal projection onto the kernel of a finite Hilbert-space operator, an instance of the known Moore–Penrose characterization of the Hyers–Ulam constant [Huang and Moslehian, 2012]. $\square$

Under affine Hilbert-space gauges $\phi_b(x) = Q_b x + g_b$, with Q_b orthogonal,

$$A'_e = Q_{t(e)} A_e Q_{s(e)}^{-1}, \quad a'_e = Q_{t(e)} a_e + g_{t(e)} - A'_e g_{s(e)}.$$

Writing Q_E, Q_C for the induced block-orthogonal maps, the linear relation operators satisfy

$$D_{A'} Q_E = Q_C D_A.$$

The translation contribution is an A' -coboundary in $\ker D_{A'}$, so also $D_{A'} a' = Q_C D_A a$. Thus singular values, residual norm, and distance to the affine exact set are gauge invariant in this sector. Indeed, for any path $p : s \rightarrow t$,

$$B'_p = Q_t B_p + g_t - A'_p g_s.$$

The final two terms cancel between relation sides because they have common endpoints and relation-exact linear parts.

The weighted norms declared by the presentation are

$$\|a\|_E^2 = \sum_e w_1(e) \|a_e\|^2, \quad \|c\|_C^2 = \sum_\sigma w_2(\sigma) \|c_\sigma\|^2.$$

In ordinary Euclidean coordinates, their normalized operator is

$$\tilde{D}_A = W_C^{1/2} D_A W_E^{-1/2}.$$

Duplicating a relation while splitting its original cell weight among the copies leaves the normalized Gram operator $\tilde{D}_A^* \tilde{D}_A$ unchanged. More general presentation refinement is not solved here. The conditioning belongs to the weighted operational presentation, not to the quotient category alone.

Proposition 7.2 (growth obstruction). *Let K_N have $N+1$ parallel edges and relations identifying adjacent edges. For noninvertible constant responses $* \mapsto i\varepsilon$, every local defect is ε , but the nearest total exact constant response in uniform edge distance is $N\varepsilon/2$ away. The relation matrix has*

$$\sigma_{\min}^+ = 2 \sin \left(\frac{\pi}{2(N+1)} \right).$$

Proof. Exactness forces all edge values to equal a common c . The smallest interval centered at c containing $\{0, \varepsilon, \dots, N\varepsilon\}$ has radius $N\varepsilon/2$. The singular values follow from the path-graph Laplacian. $\square$

This is a changing-presentation conditioning example, not a fixed-source Ulam-instability theorem.

8 Amplitude-calibrated response order

Scaling claims require a physical intervention amplitude. A coordinate name such as ε is not enough.

Definition 8.1 (response order). A cell germ is a family (σ_τ, x_τ) with declared physical amplitude $\tau \downarrow 0$. Admissible reparameterizations are cofinal increasing homeomorphisms $\tau' = \phi(\tau)$ with

$c\tau \leq \phi(\tau) \leq C\tau$ near zero. Fix reference units $\Delta_0, \tau_0 > 0$. Let S be the cofinal set of scales on which both routes succeed with finite value defect $\Delta(\tau)$. Define

$$\text{ord } \Delta = \liminf_{\substack{\tau \downarrow 0 \\ \tau \in S}} \frac{\log(\Delta(\tau)/\Delta_0)}{\log(\tau/\tau_0)},$$

using $\log 0 = -\infty$. Eventual exact agreement has order $+\infty$. Record the failure-scale set

$$\mathcal{F} = \{\tau : \text{exactly one route succeeds at scale } \tau\}$$

separately, distinguishing accumulation of failures at zero from eventual failure. If successful scales are not cofinal, no value order is assigned.

Proposition 8.2 (response-order calculus). *1. Response order is invariant under uniformly bi-Lipschitz fiber gauges and amplitude reparameterizations.*

2. If a uniformly Lipschitz continuation is defined on both route outcomes for all sufficiently small $\tau \in S$, it cannot decrease response order when both orders are computed on that same tail of S .

3. Under the same common-scale hypothesis, a uniformly bi-Lipschitz continuation preserves response order.

Proof. Multiplicative constants contribute bounded logarithmic terms, which vanish after division by $\log \tau$. The continuation claims follow from the one-sided and two-sided Lipschitz inequalities on the same scale set. A partial continuation that deletes a cofinal subsequence can change a liminf, so no claim is made without this hypothesis. $\square$

The order separates familiar regimes when amplitude is calibrated:

Table 2: Response-order signatures under uniform regularity.

Mechanism	Defect scaling	Response order
smooth connection, nonzero curvature	$\Theta(\tau^2)$	2
transverse active-set projection example	$\Theta(\tau)$	1
fixed jump crossed by shrinking cells	$\Theta(1)$	0
one-sided route feasibility	hard failure	nonempty $\mathcal{F}$
exact response	0	$+\infty$

For a fixed C^3 connection in a common chart, with uniformly bounded derivatives, $x_\tau \rightarrow x_0$, and a nonzero limiting curvature coefficient in the chosen directions, two-order square transport has vector defect

$$\tau^2 \Omega(u, v) + O(\tau^3)$$

after choosing a local vertical identification. This is ordinary curvature theory. For

$$A_\tau = \{x \geq \tau\}, \quad B_\tau = \{x + y \geq \tau\},$$

sequential Euclidean projections from the origin end at $(\tau, 0)$ and $(\tau, \tau/2)$, giving exact defect $\tau/2$. A response of order 1 or 0, or with failures accumulating at zero, cannot be represented to leading order by one fixed uniformly regular smooth connection.

9 Worked applications

9.1 Configuration repair

Let $z = (x, y) \in \mathbb{R}_+^2$ encode two configuration capacities and let $\lambda > 0$. Two independently deployable policies are

$$A : x \geq \lambda, \quad B : x + y \geq 2\lambda.$$

At each vertex of [figure 1](#), the fiber is the cumulative feasible set. Every carry edge is Euclidean projection onto its target fiber. From $z_0 = 0$,

$$\begin{aligned} A \text{ then } B : 0 &\mapsto (\lambda, 0) \mapsto (3\lambda/2, \lambda/2), \\ B \text{ then } A : 0 &\mapsto (\lambda, \lambda) \mapsto (\lambda, \lambda). \end{aligned}$$

The value defect is

$$\frac{\lambda}{\sqrt{2}}.$$

For a point mass at the origin and truncation scale $M = \lambda$, the full empirical signature is

$$(A_\mu^+, A_\mu^-, C_\mu, V_{\mu, M}) = (0, 0, 0, 1/\sqrt{2}).$$

A reset from the baseline projects directly to (λ, λ) under either history.

Under the assumed homogeneous policy family, this is a closed-form extrapolation rather than independent field evidence. A future holdout test could calibrate the two single-policy projections and Euclidean edit metric at $\lambda = 1$, then withhold $\lambda = 2$. The predicted endpoints are $(3, 1)$ and $(2, 2)$, with defect $\sqrt{2}$. A different result would falsify the edit metric, homogeneous scaling, or carry implementation. The configuration label is appropriate only when the controllable capacities and quadratic edit cost are credible; discrete configuration systems require different fibers and maps.

Now append a validation intervention C whose response is the identity on the partial domain

$$D_C = \{(x, y) : y \geq 3\lambda/4\}.$$

The A -then- B endpoint has $y = \lambda/2$ and fails C ; the B -then- A endpoint has $y = \lambda$ and passes. Both endpoints are $\lambda/4$ from the guard, less than their mutual distance $\lambda/\sqrt{2}$. Thus the imported guard lemma correctly localizes the conversion from finite value debt to hard failure. Here the guard is constructed as a demonstration. In a prospective test it must be declared before observing the two-path outcomes.

9.2 Online allocation with hard failure

There are two unit-capacity servers. Flexible job A prefers server 1 but can use either; rigid job B can use only server 1. Carry response is nonpreemptive greedy admission. The final demand set $\{A, B\}$ is externally independent of arrival order.

Starting empty,

$$A \text{ then } B : \quad \emptyset \mapsto (A \rightarrow 1) \mapsto \text{failure},$$

while

$$B \text{ then } A : \quad \emptyset \mapsto (B \rightarrow 1) \mapsto (B \rightarrow 1, A \rightarrow 2).$$

The second edge of the first route is undefined. The response signature therefore has route-asymmetric feasibility, not merely a large scalar value error. A reset scheduler allowed to migrate incumbents reaches $(B \rightarrow 1, A \rightarrow 2)$ from either history.

For completeness, give every fiber the unit discrete metric and set

$$X_\emptyset = \{\emptyset\}, \quad X_A = \{A \rightarrow 1, A \rightarrow 2\}, \quad X_B = \{B \rightarrow 1\}, \quad X_{AB} = \{(B \rightarrow 1, A \rightarrow 2)\}.$$

The four edge maps send $\emptyset \mapsto A \rightarrow 1$, $\emptyset \mapsto B \rightarrow 1$, $A \rightarrow 2 \mapsto (B \rightarrow 1, A \rightarrow 2)$ with $A \rightarrow 1$ outside the domain, and $B \rightarrow 1 \mapsto (B \rightarrow 1, A \rightarrow 2)$. They are partial nonexpansive maps. At the empty input, with $p = A\text{-then-}B$ and $q = B\text{-then-}A$, the directional signature is $A_\mu^+ = 0$, $A_\mu^- = 1$, and $C_\mu = 0$.

Preferences, compatibility, and nonpreemption are protocol assumptions here. A prospective calibration would require forced-unavailability trials for compatibility and sequential trials for nonpreemption, with the target pair withheld. The resulting order-specific admission consequence is then falsifiable. An explicit reservation policy that assigns A to server 2, or a migration rule that moves A there when B arrives, makes both orders succeed.

9.3 Smooth constrained transport

For comparison, consider internal coordinates (x, y, z) , external coordinates $b = (b_1, b_2)$, and the regular constraint

$$z - b_1x - b_2y = 0.$$

Minimum Euclidean internal velocity gives, with $D = 1 + b_1^2 + b_2^2$, horizontal fields

$$\begin{aligned} h_1 &= \partial_{b_1} - \frac{b_1x}{D} \partial_x - \frac{b_2x}{D} \partial_y, \\ h_2 &= \partial_{b_2} - \frac{b_1y}{D} \partial_x - \frac{b_2y}{D} \partial_y. \end{aligned}$$

At $b = 0$,

$$[h_1, h_2] = -y \partial_x + x \partial_y.$$

For initial $(x, y) = (1, 2)$, the predicted normalized two-order difference is $(-2, 1)$. Numerical transport gives the convergence in [table 3](#).

Table 3: Smooth two-order defect divided by τ^2 .

τ	first component	second component	error to $(-2, 1)$
0.2000	-1.90528	0.97987	0.09684
0.1000	-1.97534	0.99499	0.02516
0.0500	-1.99377	0.99875	0.00635
0.0250	-1.99844	0.99969	0.00159
0.0125	-1.99961	0.99992	0.00040

This numerical example checks the smooth reduction; it is not evidence that connection curvature is an NCS invention.

10 A modeling and falsification protocol

An NCS model should be built and tested in the following order.

1. **Declare external equivalences.** Attach a relation cell only when the two paths genuinely represent the same net external intervention. If timing or deadlines are part of the intervention, arrival orders may not be equivalent.
2. **Specify state fibers and selection rules.** State whether response is reset, carry, projection, optimization, greedy admission, or another mechanism. Resolve nonunique selections explicitly.
3. **Calibrate metrics and moduli.** A defect magnitude is meaningless without units, a cost metric, and Lipschitz/amplification information.
4. **Estimate the full signature.** Report both directions of route-asymmetric failure, common failure, and finite value defect.
5. **Predict held-out compositions.** Fit elementary response maps or local cells; withhold longer paths or new amplitudes.
6. **Probe scale.** Use a physically calibrated intervention amplitude to distinguish smooth, seam, jump, and hard-failure behavior.
7. **Compare with reset.** Full relaxation should remove carry residue when a unique reset state exists. If it does not, the model has conflated semantics.
8. **Test rectifiability.** In a tractable response sector, estimate whether small local residuals imply a nearby exact protocol or are amplified by presentation conditioning.

11 Research agenda

The proposed branch becomes valuable only if it produces more than a common vocabulary. The following problems are concrete.

Partial Ulam stability. Characterize presentations and target classes for which small value defects, small domain symmetric differences, and low common failure imply proximity to an exact partial response functor. Existing group/operator stability is the benchmark, not the answer.

Domain-boundary geometry. Replace the elementary seam margin by sharp distributional bounds using boundary regularity, transversality, reach, or metric-measure concentration. Determine when cell-level seam exposure controls longer path failure.

Quantitative presentation comparison. Develop operationally meaningful weights and comparison theorems under subdivision or Tietze moves. The response presentation is physical data, but purely notational refinement should not change predictions.

Set-valued and stochastic response. Nonunique repairs require correspondences, kernels, or distributions. Directed Hausdorff excess, symmetric Hausdorff distance, and optimal transport encode different semantics; no one choice is canonical.

Stratified comparison. Build rigorous comparison theorems among smooth curvature, active-set projections, saltation products, and junction-order defects without asserting a nonexistent general additive decomposition.

Inference and experimental design. Design minimal collections of open paths and exchange cells that identify held-out order signatures. Determine sample complexity for seam exposure and response order.

Protocol design. Optimize adjustment costs, tie-breakers, reservation policies, or migration rules to reduce asymmetric failure while respecting intervention cost.

Natural review communities. The proposal is intentionally legible to existing specialties. Category and rewriting theorists can assess presentations, fillings, and stability; variational analysts and hybrid-systems researchers can assess partial response and guards; operations researchers and online-algorithm specialists can assess allocation protocols; mathematical economists can assess the comparative-statics semantics. A credible review should include at least one formal-structure reviewer and one application/protocol reviewer.

12 Limitations and field-status assessment

The framework has six immediate limitations.

1. The intervention presentation and its weights are model choices. Different choices can change conditioning and even which paths are compared.
2. A state metric may be physical, economic, normative, or analyst-chosen. It must transform covariantly under units and be justified in applications.
3. Deterministic selection hides nonunique repair. The present theory makes undefined selection explicit but does not yet solve the set-valued case.
4. Scalar value defects discard direction and nonabelian operator information. In the smooth sector, vector or operator curvature is primary.
5. The literature map is extensive but not exhaustive. “No mapped source” is not a global priority proof.
6. The present positive results are reductions to established theories or elementary composition bounds. No theorem here yet forces recognition of an independent mathematical branch.

Does this justify a new branch? It justifies a *candidate research program*. The recurring modeling target is real, the object is precise, the guard-robustness baseline becomes operational in response composition, and the same signature transfers across continuous repair and partial allocation. It would be premature to claim that the program has reached the maturity of graph theory, information geometry, or variational analysis. Field status should depend on stronger partial-rectification theorems, cross-domain predictive successes, and independent peer evaluation.

The name “noncommutative comparative statics” is intended literally: externally equivalent parameter changes need not have commuting state responses. It does not refer to operator-algebraic noncommutative geometry. The subtitle should accompany the name until the usage is stable.

13 Conclusion

The central object of NCS is not an equilibrium map and not a connection alone. It is a stateful response assignment on a presented space of external interventions. Its basic observable is not one number but a signature: which order succeeds, whether both fail, and how successful outcomes differ. Imported signed-distance robustness shows where finite order debt can become hard feasibility asymmetry. The response-order convention compares regular curvature with active-set and jump behavior once intervention amplitude is calibrated. Hyers–Ulam rectification, robust margins, rewriting bounds, projection theory, and holonomy enter as established sectors.

This reframing is useful because it converts a familiar but scattered motif – locally reasonable repairs that compose badly – into a falsifiable mathematical program. The next decisive test is not another name or analogy. It is whether partial-domain stability and held-out order prediction yield theorems and measurements that the neighboring fields do not already provide in the same form.

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